

Environmental Data Science: Homework #5
Univariate and Bivariate Distributions
Due: Before Class on 9/30/26

Plotting Practice

Using the EPA AQS data make the following plots, and provide a 1-2 sentence interpretation of what they show. Make sure your figures follow the best practices described in class.

- i. A histogram of the O₃ concentrations
- ii. A density plot of NO₂ concentrations
- iii. A scatter plot of O₃ vs NO₂ concentrations*
- iv. A barchart of the number of days per month with O₃ > 35 ppb

*Merging datasets can be difficult. Let's say you have two vectors of observations, each with an associated date vector. A challenge in environmental data is *missing data*, or *gaps*. If these are present, you can't just put both observation vectors into a data frame with the same date, because they might be different lengths!

So, if you have a vector V1 with associated dates date1_vector and a vector V2 with associated dates date2_vector, you can **merge** the two, combining so that missing data is automatically removed. This might help:

```
df_1 = pd.DataFrame(data = {"V1_Name" : V1, 'date' : date1_vector})
```

```
df_2 = pd.DataFrame(data = {"V2_Name" : V2, 'date' : date2_vector})
```

```
df_all = df_1.merge(df_2, on=['date'])
```

Mauna Loa CO₂

Download the Mauna Loa CO₂ data from Brightspace.

- i. Make a scatterplot of average CO₂ concentrations vs time (decimal date).
- ii. Make the worst possible scatterplot of average CO₂ concentrations vs time (decimal date).
 - a. Make the figure *technically* correct (it has to show CO₂ concentrations vs time), but horrible

Submit your code via GitHub along with a PDF submitted to Brightspace.